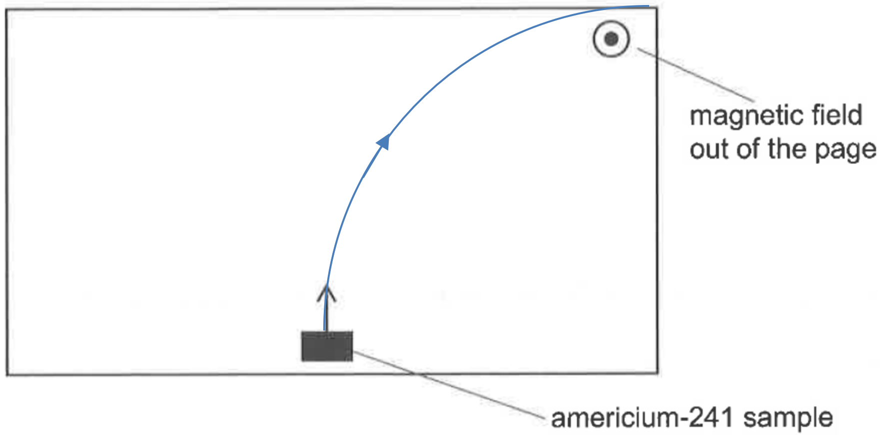
	Therefore it will be safer to use a cable of several thin wires.
5(a)	$\frac{1}{2}\left(6.64\times10^{-27}\right)v_{\alpha}{}^{2}=8.78\times10^{-13}$ $v_{\alpha}=1.6262\times10^{7}\text{ m s}^{-1}$ $v_{\alpha}\approx1.63\times10^{7}\text{ m s}^{-1}$
5(b)(i)	

No.	Solution
5(b)(ii)	Magnetic force on α-particle $=(0.682)\left(2\times1.60\times10^{-19}\right)\left(1.6262\times10^{7}\right)$ $=3.549\times10^{-12}$ N $\approx 3.55\times10^{-12}$ N
5(c)	Kinetic energy of α-particle decreases due to work done against air resistance. Speed of α-particle decreases. Therefore the radius of the circular path decreases.
5(d)	1. A β-particle is negatively-charged. Using Fleming's Left Hand Rule, the β-particle will be deflected leftwards. It will follow a circular path to the left. 2. Comparing mass-to-charge ratio, $\left(\dfrac{m}{q}\right)_\alpha : \left(\dfrac{m}{q}\right)_\beta \approx 3644$. Comparing speed, $v_\alpha : v_\beta \approx 0.1$. Using $r=\left(\dfrac{m}{q}\right)\left(\dfrac{v}{B}\right)$, the radius of the circular path of β-particle is much smaller than that of an α-particle.
6(a)(i)	The vertical velocity of the ball decreases at a constant rate as the ball is moving